soon becomes clear, however, that inside the bottle many physical entities remain: electric and magnetic fields, and a constant swarming of quantum particles. The existence of a complete void, without any physical entity except amorphous space, “absolute, true, and mathematical,” remains a brilliant theoretical idea introduced by Newton to found his physics on, for there is no scientific, experimental evidence to support its existence. An ingenious hypothesis, perhaps the most profound insight achieved by one of the greatest scientists—but is it one that actually corresponds to the reality of things? Does Newton’s space really exist? If it exists, is it really amorphous? Can a place exist where nothing exists?

The question is identical to the analogous one regarding time: does Newton’s “absolute, true, and mathematical” time exist, flowing when nothing happens? If it exists, is it something altogether different from the things of this world? Is it so very independent from them?

The answer to all these questions lies in an unexpected synthesis of the apparently contradictory ideas held by these two giants. And to accomplish this, it was necessary for a third giant to enter the dance.*

THE DANCE OF THE THREE GIANTS

The synthesis between Aristotle’s time and Newton’s is the most valuable achievement made by Einstein. It is the crowning jewel of his thought.

The answer is that the time and space Newton had intuited the existence of, beyond tangible matter, do effectively *exist*. They are real. Time and space are real phenomena. But they are in no way absolute; they are not at all independent from what happens; they are not as different from the other substances of the world, as Newton had imagined them to be. We can think of a great Newtonian canvas on which the story of the world is drawn. But this canvas is made of the same stuff that everything else in the world is made of, the same substance that constitutes stone, light, and air: it is made of fields.

Physicists call “fields” the substances that, to the best of our knowledge, constitute the weave of the physical reality of the world. Sometimes they may be given exotic names: the fields “of Dirac” are the fabric of which tables and stars are made. The “electromagnetic” field is the weave of which light is made, as well as the origin of the forces that make electric motors turn and the needle of a compass point north. But—here is the key point—there is also a “gravitational” field: it is the origin of the force of gravity, but it is also the texture that forms Newton’s space and time, the fabric on which the rest of the world is drawn. Clocks are mechanisms that measure its extension. The